

Julian Garcia-Abadillo, PhD

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Google Scholar | ResearchGate | ORCID | Github | LinkedIn

Research Interests

My research focuses on applying statistical methods to agricultural problems, mostly within the plant breeding field. I am particularly interested in using quantitative genetics and its statistical foundation for trait prediction in academic research, paired with the industry machine learning paradigm and its focus on real-world applicability and scalability.

Current lines of my research explore multi-omics datasets (including genomic, pedigree, weather, and phenomic records) for phenotypic prediction and genotype-by-environment pattern identification. Additionally, I work on developing methodologies to select optimal information sources for training better models and implementing non-linear procedures to predict dynamic, time-related traits.

Keywords: *Quantitative Genetics, Plant Breeding, Statistics, Data Science, R and python programming, Genomic Selection, Genome-wide Association Studies (GWAS), Machine Learning, Artificial Intelligence, Data Visualization, Genotype-by-Environment Interactions, High-Throughput Computing, Multi-omics Integration*

Education

PhD. Quantitative Genetics

09/2021 – 12/2025

Technical University of Madrid (UPM) – Center for Plant Biotechnology and Genomics (CBGP)

PhD Advisor: Dr. Julio Isidro y Sánchez

Dissertation: *Machine Learning Approaches Applied to Genomic-Assisted Plant Breeding*

MSc. Artificial Intelligence

09/2020 – 07/2021

Technical University of Madrid (UPM) – School of Computer Engineering (ETSIInf)

Thesis Advisors: Dr. Alfonso Mateos Caballero & Dr. Antonio Jiménez Martín

Thesis: *Modeling the Propagation of a Disease Using Graph Theory and Phone Mobility Data*

BSc. Biotechnology

09/2016 – 07/2020

Technical University of Madrid (UPM) – School of Agricultural, Food & Biosystems Engineering (ETSIAAB)

Thesis Advisors: Dr. Patricia Giraldo Carbajo & Dr. Laura Pascual Bañuls

Thesis: *Identification of Genomic Regions Involved in Wheat Root Development*

Work Experience

University of Florida

Gainesville, FL, US

Research Assistant

09/2024 – 02/2025 | 08/2025 – present

Supervisor: Dr. Diego Jarquín

My duties at UF involve providing quantitative genetics and statistical support for various projects and collaborations related with plant breeding and tree orchard management programs. Additionally, I advise MS and PhD students and serve as a TA and guest lecturer.

Google LLC

Mountain View, CA, US

AI/Plant Biology PhD Resident

01/2024 – 07/2024

Supervisor: Dr. Tim Beissinger

As a PhD Resident at Google, I reported to the technical lead of the project. I led independent research initiatives, leveraging Google's proprietary data and computational infrastructure to deliver insights for client companies. My role involved producing and reviewing python and R code for data science pipelines, providing scientific and statistical guidance to software engineers and translating research findings into summaries for leadership.

Publications

Peer-Reviewed Papers

I have authored or co-authored 15 peer-reviewed papers (3 as first author), with 4 more accepted for publication.

1. **Garcia-Abadillo**, J., Adunola, P., Aguilar, F., Trujillo-Montenegro, J., Riascos, J., Persa, R., Sanchez, J., & Jarquin, D. (2024). Sparse testing designs for optimizing predictive ability in sugarcane populations. *Frontiers in Plant Science*, 15, 1400000.
2. **Garcia-Abadillo**, J., Barba, P., Carvalho, T., Sosa-Zuniga, V., Lozano, R., Carvalho, H., Garcia-Rojas, M., Salazar, E., & Sanchez, J. (2024). Dissecting the complex genetic basis of pre-and post-harvest traits in *Vitis vinifera* L. using genome-wide association studies. *Horticulture Research*, 11(2), uhad283.
3. **Garcia-Abadillo**, J., Morales, L., Buerstmayr, H., Michel, S., Lillemo, M., Holzapfel, J., Hartl, L., Akdemir, D., Carvalho, H., & Isidro-Sanchez, J. (2023). Alternative scoring methods of fusarium head blight resistance for genomic assisted breeding. *Frontiers in Plant Science*, 13, 1057914.
4. Caballero, E., **Garcia-Abadillo**, J., & Jarquin, D. (2026). Missing comparability: When genomic selection faces field variability. A case study in soybeans. *The Plant Genome*, 19(2), e70264.
5. Proma, S., **Garcia-Abadillo**, J., Sagae, V., Sacks, E., Leakey, A. D., Zhao, H., ... & Jarquin, D. (2026). Multi-trait multi-environment genomic prediction strategies for *Miscanthus sacchariflorus*. *Frontiers in Plant Science*, 17, 1835247.
6. Kaur, S., Sleper, J., Dalid, C., **Garcia-Abadillo**, J., Whitaker, V. M., & Jarquin, D. (2026). Modeling production curves in a strawberry breeding program to optimize early season productivity. *Frontiers in Plant Science*, 17, 1808529.
7. Proma, S., **Garcia-Abadillo**, J., Shaik, A., & Jarquin, D. (2026). Optimizing genomic prediction-based sparse testing designs for multi-environment trials in tetraploid potato cultivars. *Crop Science*, 66(2), e70260.
8. Fritsche-Neto, R., Resende, R. T., Olivoto, T., **Garcia-Abadillo**, J., Nascimento, M., Bahia, M. A. M., ... & Vieira, R. A. (2025). Prediction-based breeding: Modern tools to optimize and reshape programs. *Crop Science*, 65(5), e70175.
9. Menor de Gaspar, J., Domínguez Rondón, A., **García-Abadillo**, J., Knox, R., Bokore, F. E., Boyle, K., ... & Isidro y Sánchez, J. (2025). Mapping novel yellow and leaf rust loci and predicting resistance in cross derived Canadian durum wheat. *The Plant Genome*, 18(4), e70124.
10. Evangelista, J. S. P. C., Dias, K. O. D. G., Pastina, M. M., Chaves, S., Guimarães, L. J. M., Hidalgo, J., **Garcia-Abadillo**, J., et al., & Jarquin, D. (2025). Optimizing the single-step model for predicting fumonisins resistance in maize hybrids accounting for the genotype-by-environment interaction. *Frontiers in Genetics*, 16, 1475452.
11. Messina, C., **Garcia-Abadillo**, J., Powell, O., Tomura, S., Zare, A., Ganapathysubramanian, B., & Cooper, M. (2025). Toward a general framework for AI-enabled prediction in crop improvement. *Theoretical and Applied Genetics*, 138(7), 1–15.
12. Gonzalez, F., **Garcia-Abadillo**, J., & Jarquin, D. (2025). Introducing CHiDO—A No Code Genomic Prediction software implementation for the characterization and integration of driven omics. *The Plant Genome*, 18(1), e20519.
13. Carvalho, H., Rio, S., **Garcia-Abadillo**, J., & Sanchez, J. (2024). Revisiting superiority and stability metrics of cultivar performances using genomic data: derivations of new estimators. *Plant Methods*, 20(1), 85.
14. Din, A., Gul, R., Khan, H., **Garcia-Abadillo**, J., Persa, R., Sanchez, J., & Jarquin, D. (2024). Assessing the Genotype-by-Environment GxE Interaction in Desi Chickpea via the Bayesian Additive Main Effects and Multiplicative Interaction Model. *Agriculture*, 14(2), 215.
15. Lopez-Fernandez, M., **Garcia-Abadillo**, J., Uauy, C., Ruiz, M., Giraldo, P., & Pascual, L. (2023). Genome wide association in Spanish bread wheat landraces identifies six key genomic regions that consti-

tute potential targets for improving grain yield related traits. *Theoretical and Applied Genetics*, 136(12), 244.

Accepted

- **Garcia-Abadillo**, J., Isidro Sánchez, J., Jarquin, D., Galeano, E., & Thomas, B. R. (in press). Prediction of conelet mortality rates in interior lodgepole pine using phenomic, nutritional, and environmental data. *Frontiers in Plant Science*, 17, 1845181.
- Martin Zapien-Macias, J., **Garcia-Abadillo**, J., Munoz, P. R., & Nunez, G. H. (in press). Agronomic Strategies to Improve Canopy Light Interception and Fruit Ripening Concentration in Southern Highbush Blueberry in North-Central Florida. *International Journal of Fruit Science*.
- Zhou, X., **Garcia-Abadillo**, J., Jarquin, D. (in press). Development of Conditional Cores Via Super Saturated Designs for Genetic Diversity Based Prediction Models. *The Plant Genome*.
- Acharya, S. R., **Garcia-Abadillo**, J., Lyster, J., Brown-Guedira, G., Jarquin, D., & Bandillo, N. (in press). Reaction Norm Modeling of High-Dimensional Genomic and Environmental Data Improves Prediction Accuracy in Winter Wheat. *The Plant Genome*.

Book Chapters

I have contributed to 2 book chapters.

1. Alemu, A., Montesinos-Lopez, O. A., Sagae, V. S., **Garcia-Abadillo**, J., Nascimento, M., Jarquin, D., & Ortiz, R. (2026). Genomic Selection Role to Mitigate G×E Interaction in Plant Breeding. *Genotype × Environment Interactions and its Implications for Plant Breeding*, 343-364.
2. Ellur, V., Ambati, S., Jarquin, D., **Garcia-Abadillo**, J., & Mural, R. V. (2026). Emerging Approaches to Sustainable Agriculture and Food Security. *Next Generation Crop Protection for Agricultural Sustainability and Food Security*, 44-62.

Manuscripts under revision

1. Arcos Jaramillo, A., **Garcia-Abadillo**, J., Murray, J., Manzanero, B., Mata, L. A., Jarquin, D., Sandoya, G. (n.d.). Seasonal and soil variations influence genotype-by-environment interactions for phosphorus use efficiency in lettuce. *The Plant Genome*.
2. Garapati, R., **Garcia-Abadillo**, J., & Jarquin, D. (n.d.). Hyper-parameter Optimization of Deep Learning for Genomic Prediction: Concepts, Strategies and Inference. Under revision.
3. Barmukh, R., Garg, V., Thudi, M., Roorkiwal, M., Chitikineni, A., **Garcia-Abadillo**, J., Jarquin, D., et al., & Varshney, R. (n.d.). Harnessing Genebank Diversity for Chickpea Improvement through Genomics and AI-Guided Analysis of 10,225 Genomes. Under revision.
4. Adaskaveg, J. A., Hershberger, J., Farmer, A., Penmetsa, R. V., Garcia-Lopez, I., **Garcia-Abadillo**, J., ... & Diepenbrock, C. H. (n.d.). Genotypic and multi-environment phenotypic evaluation of the lima bean USDA National Plant Germplasm System collection. Under revision.

Manuscripts in preparation

1. **Garcia-Abadillo**, J., Barba, P., & Isidro y Sánchez, J. (n.d.). Performance of Genomic Selection Models and Cross-Validation Strategies for Diverse Traits in Stratified Grapevine Populations. Manuscript in preparation.

Scientific Communications

Talks

I have delivered 6 invited/conference talks (5 conference presentations and 1 keynote address).

1. *The importance of data quality*. Conference on Applied Statistics in Agriculture and Natural Resources, Gainesville, FL, US. **May 11, 2025**
2. *Integration of multi-omics for phenotypic prediction in plant breeding*. XXXII Plant and Animal Genome Conference, San Diego, CA, US. **Jan 10, 2025**

3. *Phenotypic prediction of abortion rate in lodgepole pine using non-genetic multi-omics datasets*. Alberta Forestry Industry Meeting, Edmonton, AB, Canada. **Nov 16, 2024**
4. *Integrating machine learning in the plant breeding toolbox: Benefits and challenges*. ASA, CSSA & SSSA International Annual Meetings, San Antonio, TX, US. **Nov 11, 2024**
- Integration of multi-omics and other interactions in predictive models* [Keynote address]. **Oct 02, 2024**
5. VIII International Meeting on Plant Breeding: AI for Scientific Breakthrough: Applications and Innovations, Piracicaba, Brazil.
6. *Integration of multi-omic datasets and their interactions into predictive models: A tree species example*. II GxExM Symposium, Gainesville, FL, US. **Nov 07, 2023**

Poster Sessions

I have presented 4 posters at scientific conferences.

- Beyond Cross-Validation in Genomic Selection: Extracting Insights from Non-Validatable Forward Predictions with CHiDO*. National Association of Plant Breeders conference, College Station, TX, USA. **Jun 15, 2026**
- AgroPower. Monte Carlo Power Analysis for Complex Agronomic Experimental Designs*. 10th Annual University of Florida Plant Science Symposium, Gainesville, FL, USA. **Jan 20, 2026**
- Predicting without genomics using CHiDO: A multi-omics case study in trees*. National Association of Plant Breeders conference, Kona, HI, USA. **May 19, 2025**
- Evaluation of Fusarium Head Blight resistance by alternative methods for genomic assisted breeding*. XVIIIth Eucarpia Biometrics in Plant Breeding Conference, Paris-Saclay University, Gif-sur-Yvette, France. **May 11, 2022**

Funding and Academic Awards

• Google University Program Relocation Stipend	2024	12,000 USD
• Complementary Visiting Scholar Fundings	2023	3,077 EUR (Approx. 3,500 USD)
• Santander-UPM Visiting Scholar Fundings	2023	4,900 EUR (Approx. 5,600 USD)
• Santander-UPM Q1 Article Publishing Fee Fundings	2023	3,000 EUR (Approx. 3,400 USD)
• Santander-UPM 4-year PhD Fundings	2022	75,000 EUR (Approx. 86,000 USD)
• Spanish Government's Undergraduate Research Stipend	2020	2,000 EUR (Approx. 2,300 USD)
• Community of Madrid Academic Excellence Award	2019	2,100 EUR (Approx. 2,400 USD)
• Community of Madrid Academic Excellence Award	2018	2,100 EUR (Approx. 2,400 USD)
• Community of Madrid Academic Excellence Award	2017	2,100 EUR (Approx. 2,400 USD)

Travel Awards

• National Association of Plant Breeders Conference <i>Agronomy Department (University of Florida)</i>	2025	Kailua-Kona, HI, US
• Conf. on Applied Statistics in Agriculture and Natural Resources <i>Agronomy Department (University of Florida)</i>	2025	Gainesville, FL, US
• 32nd Plant & Animal Genome Conference <i>Edwardson Agronomy Endowment (University of Florida)</i>	2025	San Diego, CA, US
• ASA-CSSA-SSSA International Annual Meeting <i>Agronomy Department (University of Florida)</i>	2024	San Antonio, TX, US
• 8th International Meeting on Plant Breeding <i>Corteva Agriscience Plant Sciences Symposia Series</i>	2024	Piracicaba, Brazil
• Alberta Forestry Industry Meeting <i>Department of Renewable Resources (University of Alberta)</i>	2023	Edmonton, AB, Canada
• Multi-omics Integration and AI short Course at CIAT <i>Agronomy Department (University of Florida)</i>	2023	Cali, Colombia

Teaching Experience

Short Course Instructor

- **IV Multi-Omic Integration for AI Genomic Prediction Breeding** **Jul 6-10, 2026**
Malachowsky Hall for Data Science and Information Technology, Gainesville, FL, US
Main instructor: Dr. Diego Jarquin
 - G×E Prediction Models using Weather Data
 - AI Methods I: Introduction to Machine Learning and Semiparametric Methods
 - AI Methods II: Nonparametric ML methods: Deep Learning & Random Forests
 - Sparse Testing Designs
 - Multi-omics Integration using CHiDO
- **III Multi-Omic Integration for AI Genomic Prediction Breeding** **Jul 7-11, 2025**
Malachowsky Hall for Data Science and Information Technology, Gainesville, FL, US
Main instructor: Dr. Diego Jarquin
 - Introduction to R and HiperGator
 - G×E Prediction Models using Weather Data
 - AI Methods I: Introduction to Machine Learning and Semiparametric Methods
 - AI Methods II: Nonparametric ML methods: Deep Learning & Random Forests
 - Sparse Testing Designs
 - Multi-omics Integration using CHiDO
- **II Multi-Omic Integration for AI Genomic Prediction Breeding** **Jul 15-19, 2024**
Straughn IFAS Extension Professional Development Center, Gainesville, FL, US
Main instructor: Dr. Diego Jarquin
 - Introduction to R and HiperGator
 - AI Methods I: Machine Learning and Kernel Methods
 - AI Methods II: Deep Learning and its Application in Plant Breeding
- **Multi-Omic Integration Short Course** **Nov 2-3, 2023**
Centro Internacional de Agricultura Tropical (CIAT), Cali, Colombia
Main instructor: Dr. Diego Jarquin
 - AI Methods I: RKHS & Deep Learning
 - AI Methods II: Random Forests

Workshop Assistant

- **Multi-Omics Integration for Applied Prediction, Concepts, Foundations, and Strategies** **May 12, 2025**
2025 Conference on Applied Statistics in Agriculture and Natural Resources
Main instructor: Dr. Diego Jarquin

Guest Lecturer

- **AGR6932: Multi-Omic Integration for Applied Prediction Breeding** **Fall 2024**
University of Florida — Agronomy Department
Main instructor: Dr. Diego Jarquin
Two lectures on Overfitting and Model Complexity (theory and practice)

Teaching Assistant

- **AGR6932: Multi-Omic Integration for Applied Prediction Breeding** **Fall 2024**
University of Florida — Agronomy Department
Main instructor: Dr. Diego Jarquin
- **D260-203000026: Genomic-Assisted Breeding [Graduate Level]** **Fall 2022**
Technical University of Madrid (UPM), Spain — Biotechnology & Plant Biology Department

Main instructor: Dr. Julio Isidro

- **D260-205000015: Genetics [Undergraduate Level]** **Fall 2022**
Technical University of Madrid (UPM), Spain — Biotechnology & Plant Biology Department
Main instructor: Dr. Estela Gimenez

Computer Skills

- **DevOps & Version Control:** Git, GitHub
- **Computer Programming:** UNIX shell scripting, R, Python, SQL
- **Parallel Computing:** HTC/HPC platforms (HiPerGator-UF, Kairos-UPM), slurm
- **Cloud Computing:** Google Cloud Platform (Computing & Storage), VertexAI
- **Operating Systems:** Windows, macOS, Linux
- **Data Science:** scikit-learn, pandas, numpy, seaborn, tidyverse, dplyr, ggplot2
- **Web Development:** HTML, CSS, Django, Shiny
- **Desktop Editing:** Microsoft Office, LaTeX, Inkscape

Professional Development

- **Innovation and Technology Transfer Course** **Apr-May, 2024**
Centre for Biotechnology and Plant Genomics – Madrid, Spain
Main instructor: Dr. Antonio Molina
Duration: 20 hours
- **Multi-Omic Integration for AI Genomic Prediction Breeding Under Different Approaches: Past, Present and Future (Short Course)** **Jul 10-14, 2023**
University of Florida - Gainesville, FL, US
Main instructor: Dr. Diego Jarquin
Duration: 25 hours
- **Phenomic Selection (Conference Workshop)** **Sep 20, 2022**
Paris-Saclay University - Gif-sur-Yvette, France
Main instructor: Dr. Rincent Renaud
Duration: 5 hours

Software Development

- **CHiDO** (R Shiny app) — Author & Maintainer Demo | Docs
- **bayesammi** (CRAN package) — Contributor Docs

Professional Memberships

- National Association of Plant Breeders **May 2025 - Present**

Editorial Activities

Journal refereeing

- The Plant Genome **8 manuscripts**